

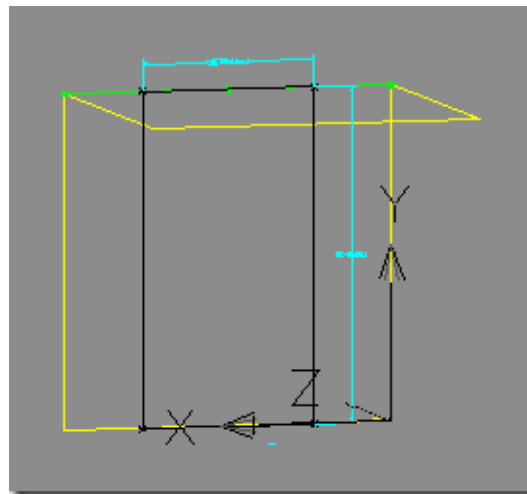
CIVIL 3D – PART BUILDER PART 3

Posted by John Evans | Dec 1, 2008 | AutoCAD Civil 3D

Source : <https://designandmotion.net/autodesk/autocad-civil-3d/civil-3d-part-builder-part-3/>

Introduction

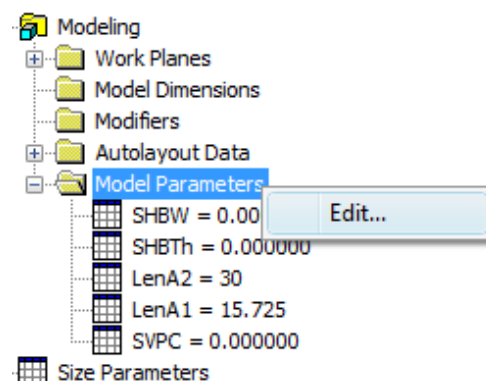
We will pick up where we left off from Part Builder Part 2. Open our saved part “Wingwall sloped” from the last session. The image below should be representative of what we had before.



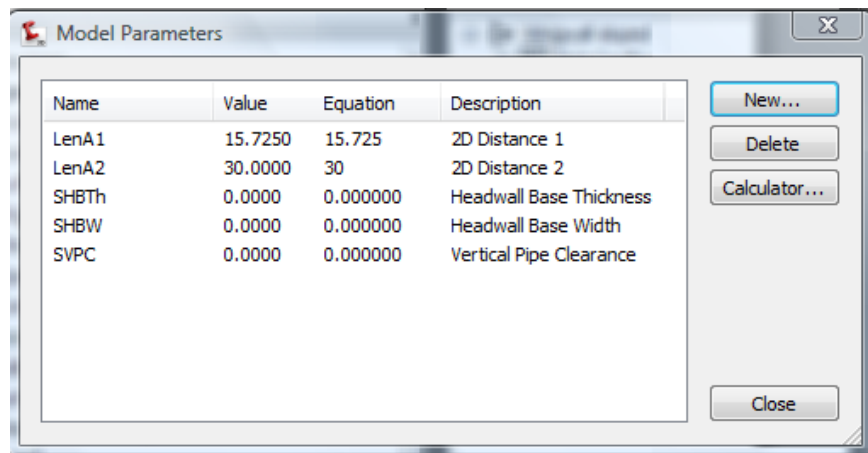
MODEL PARAMETERS

The Model Parameters are the nuts and bolts variables that control the size of the part. The parameters are either dimensions or offsets that are controlling the size and angle of features, or parameters that the user has defined. Expand the collection heading, and you should be able to see all the parameters stored in the part, as well as the values currently assigned. Lets create some user defined parameters to help us in our equations.

Right click on the Model Parameters collection, and select ‘Edit...’



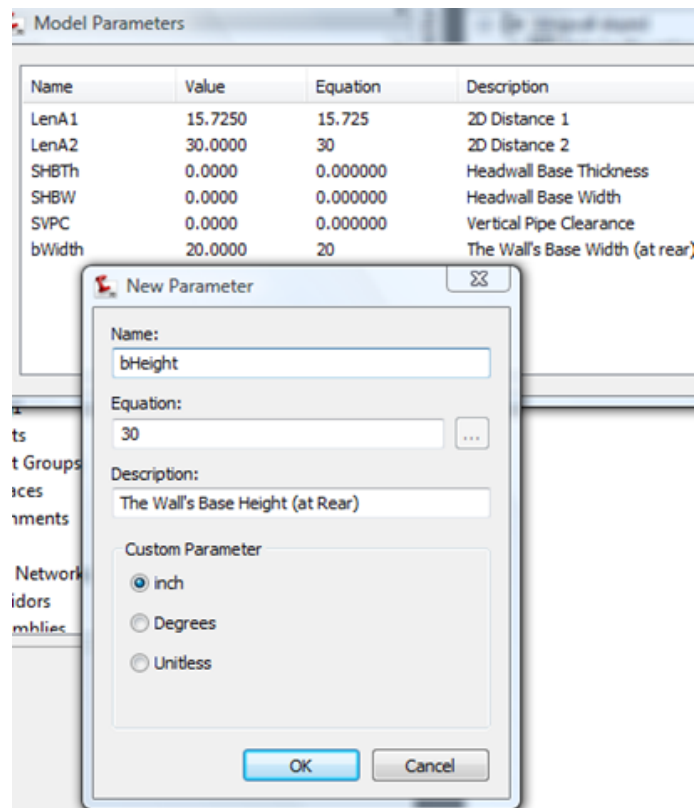
The Model Parameters dialog will appear, allowing changes, additions, and deletions to the existing parameters. On the right you will see buttons, all of which are self explanatory, with the exception of the 'Calculator...' button. This allows you to pull variables from a list (without having to remember and type them), and evaluate the result of a proposed equation. Any time you need to map a parameter to another, you can use this feature instead of typing if you wish.



Lets add 3 user parameters: bWidth, bHeight, and bThick.

Pick the New... button. The New Parameter dialog will appear, allowing our data to be configured. Enter the following data in their respective fields: 'bWidth' as the Name, 20 as the Equation, and Inch as Custom Parameter.

A description is not necessary, but sometimes helpful, especially with existing dimensions, as you will see later. If you desire a description, enter it now. Click OK.

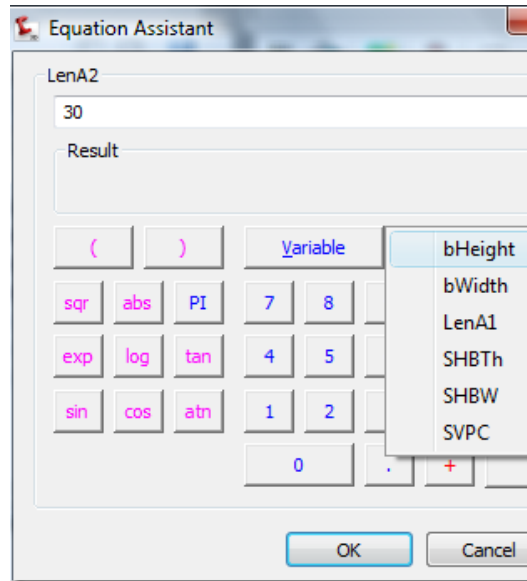


Repeat the process for bHeight, using 30 as the Equation, and bThick, using 6 as the Equation.

The Dimensions we applied in the last session are listed here as well. It would be nice to drive these, and future dimensions by a small collection of user parameters of our choosing. LenA1 is the width of the Profile feature, and LenA2 is the height.

[*TIP: You might set your dimensions in a consistent order, so that when you are in the user parameters, you can more easily distinguish between items in each set of dimensions. I try to keep the old WxHxL thing going. So LenA3 is Width, and LenA4 is Height. You should also change the description of key dimensions when you have the opportunity. When you get 30 dimensions and they all have the same automated description, it will be confusing, and you may find yourself referring to the model over and over. Change those key descriptions when you go to the Model Parameters.]

Double click on the LenA1 value under the equation column, and change the value to be 'bWidth'. Select LenA2, and click 'Calculator...'. Select Variable, and then 'bHeight' from the list. Click OK.



The LenA1 and LenA2 values will update accordingly, as well as the part features.

You should also notice the variables SHBTh (Structure Base Thickness), SHBW (Structure Base Width), and SVPC (the all important Structure Pipe Vertical Clearance) . We will change these soon, but I want everyone to have a feel for how the parameters work before we go crazy mapping everything.

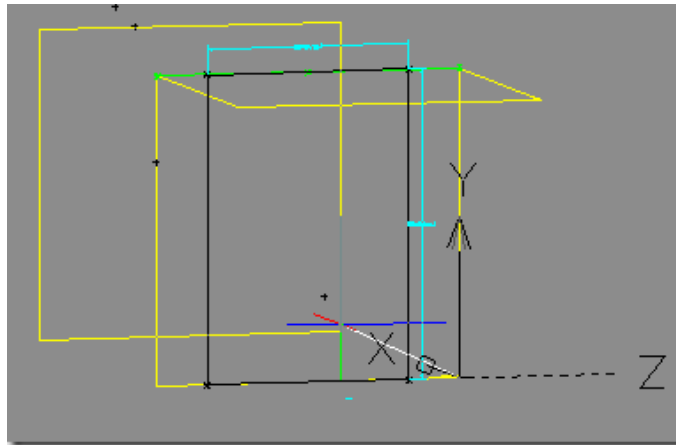
Close the Parameters Dialog.

We can now create a solid from our controlled feature. An extrusion would be the simplest, selecting a blind method that builds from the plane, away or toward us (away would have been our choice), at a specified distance. We however will have to go another route, as our main wall will not be rectangular, but instead trapezoidal. We will create a transition to help us.

TRANSITION

A transition is simply an extrusion that can taper from a profile on a start plane, to one on an end plane. To build our trapezoid, a transition will be the best choice. Subsequently, we will need to complete add an offset plane and a profile thereon. Eventually we will come back to this plane and add 2 more profiles, but I will leave that until later, to avoid confusion.

Create an offset plane, from the current 'Right Plane', away from your viewpoint, a distance of 6". Right Click the Work Planes folder and select 'Add Work Plane...'. Select 'Offset', Enter a description (I used "Rear OS") and click OK.



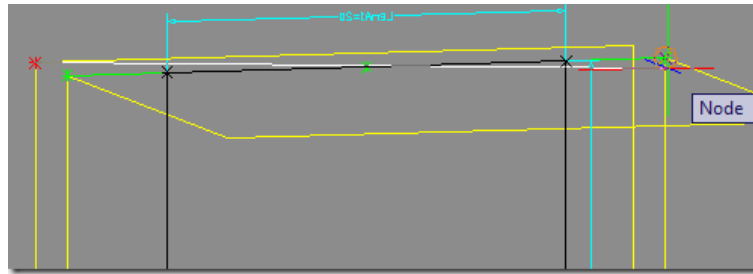
The application will prompt you to select a work plane, and to pull it in the offset direction. The white leader running from the Right Plane's ucs icon is tracking the distance you pull to apply it to the orthogonal direction. It doesn't matter where you put it, provided that it is set in the correct direction.

Model Parameters			
Name	Value	Equation	Description
bHeight	30.0000	30	The Wall's Base Height (at Rear)
bWidth	20.0000	20	The Wall's Base Width (at rear)
LenA1	20.0000	bWidth	right plane base Width
LenA2	30.0000	bHeight	right plane base Height
SHBTh	0.0000	0.000000	Headwall Base Thickness
SHBW	0.0000	0.000000	Headwall Base Width
SVPC	0.0000	0.000000	Vertical Pipe Clearance
WPOf1	6.0000	6	Workplane Offset 1

We need to set the value of the new Plane's offset distance to something we can control. Right click on the Model Parameters collection, and select 'Edit...'. Double click on the WPOf1 value under the equation column, and change the value to be '6'.

Close the dialog and the Offset Plane will update to be 6" away from the 'Right Plane'. We need to provide an anchor for the new profile. We need to repeat the same process we did when we started anchoring the 'Right Plane's content.

For the base line, we will use a reference instead of the corners of the plane.

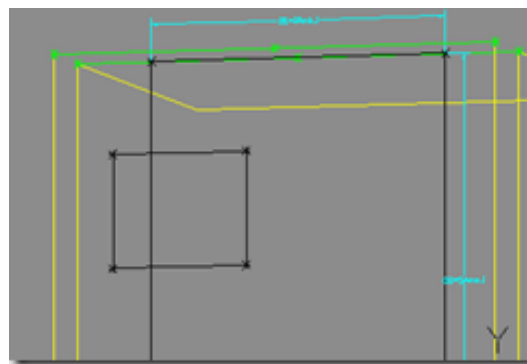


Right click on the Rear OS plane, Select Geometry, and then Line. Pick the green anchor points (using Node Osnap) from the Right Plane previously established.

The new points and line will land on the current plane, not the 'Right Plane' that we are picking. We need to have an orthogonal offset of the references on the other plane if we want out transition to line up perfectly. It won't stay like that for long, so in order to keep it there, we need to get it fixed. Right Click on the Rear OS plane, expand the Geometry collection, right click each of the points and the line, and select 'Fixed' for each item.

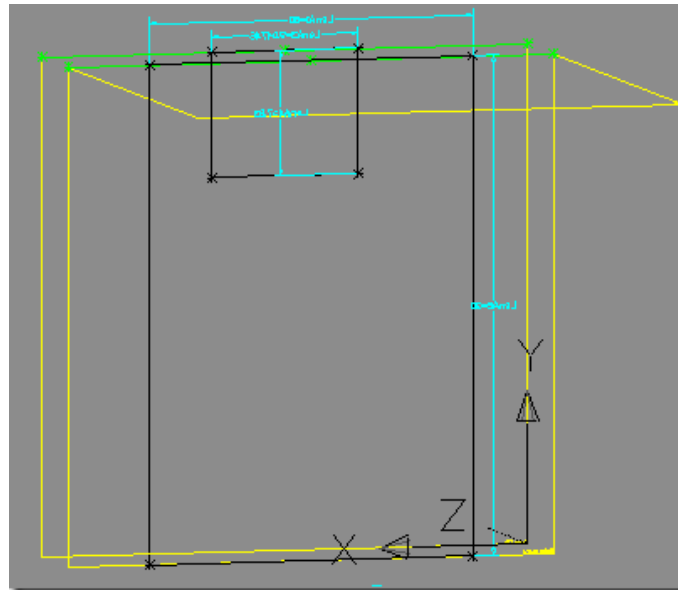
Now we can create the mid point anchor and constrain it as well. Right click on the Rear OS plane, Select Geometry, and then Point. Set it near the green line as before. Right click on the Rear OS plane, Select Constraints. Constrain the new point using the new green points on the Rear OS plane, with the Equal Distance Constraint and the Coincident Constraint, just as we did in the last session. Don't forget to fix the point as well.

Right Click on the Rear OS plane, expand the Geometry collection, right click the last point and select 'Fixed'. I know that the repeated anchoring is annoying, but it will pay off nicely in the end. Now that we have a permanent center and angular reference, we can create the profile we need. Right click on the Rear OS plane, Select 'Add Profile', and then 'Rectangular...'.



Locate the Profile in a convenient location. Remember, what size you make it will not matter until we apply the dimensions and constraints, just put it somewhere easy to get to. Now that we have the rear of the wall, let's get it into shape, and locked down. We will use the exact same procedure as in the last session:

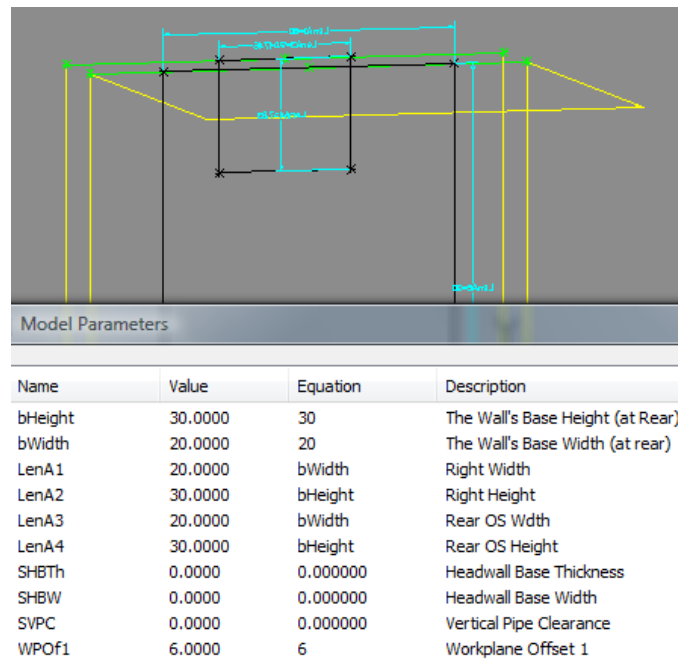
Add 2 Dimensions, Apply 'Equal Distance' constraints referencing the center green point, and then a 'Coincident' constraint referencing the top line of profile to the green anchor line. You should have something like the image below.



At this point we would probably set our dimensions to reinforce our understanding and visually that the features are sized appropriately, or we could go ahead and create the 3D solid, and later affect the parameters until the shape is what we want to see. As much as I would like to create the completely odd shaped transition now so we can see the dramatic effect the parameters will have over the part, I feel that 1/4 of the readers may get lost with the shapes being so odd, and get stuck asking 'What is this shape?'. We will actually use 1 or 2 extra steps in and out of the parameters, but this should prove to be a better way to bring everyone along a bit easier.

Navigate to the model parameters, and edit the new dimensions.

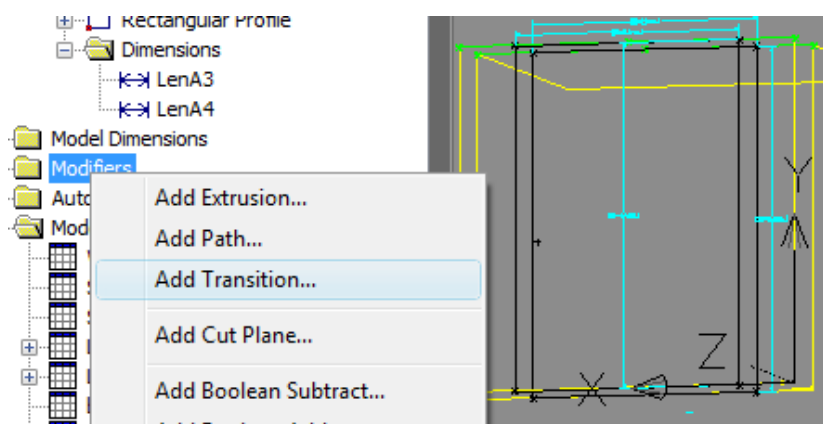
Note: We could have performed this with the dimensions in the Rear OS Plane collection, but we cannot map other parameters there, only decimal numbers, so the *Model Parameters is like one stop shopping*.



In the image above you should notice that I changed the descriptions. I needed a bit more clarity. These are for YOUR benefit, so change them however you need to be able to identify them without looking at the model. Trust me, as the model gets more complex, you need to be able to stay in the parameters and function with a clear mind.

Change the LenA3 and LenA4 to bWidth and bHeight respectively. Now the features are completely running on the two values bWidth, and bHeight. When we return to the model, it will update as indicated in the parameters.

Now on to our 3D solid modifier. Right Click 'Modifiers', and select 'Add Transition...'.

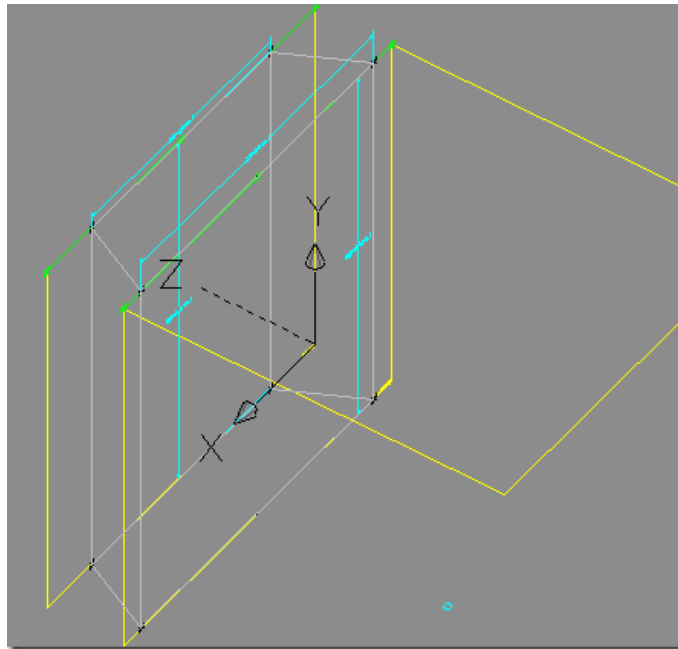


The application will prompt you to select start profile, and end profile. I have never seen that the order matters, but you should always model in a consistent manner, so that when a part goes wrong, you don't have to recreate everything to get consistency into the debugging..

Select one of the Profiles created, and then select the remaining profile. The 3D solid will be created immediately. At this point we have a rectangular headwall. The steps involved to create a simple rectangular headwall is about 1/4 the steps shown in this workflow, but that will not provide the angular transition that needed in this example.

This is the end of this session, but just for fun, let's play with the Model Parameters and see how the changes react with the model (I can never help myself, and that is why I am always late for things after modeling a new part, because I have to play with it for awhile. I'm still a kid inside).

In the Model Parameter Dialog, change the With variable LenA1 from "bWidth" to " $bWidth + bThick$ ". Close the dialog, and we see your part change like this.



Save the part. This should give you an idea that the next session will be even more enjoyable. We will add regions to the existing Planes, and Regions on new planes, with a lot of equations and referencing in the model parameters. If you are new to modeling, the next session should prove to be very satisfying.